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Assignment 2 - 11th of March 2026 ## 1.determine the final output vector Input vector x:

$$x = \begin{pmatrix} 1 \\ 0 \\ -1 \\ 2 \\ 1 \\ 0 \\ -2 \\ 1 \end{pmatrix}$$

Layer 1: First Hidden Layer (ReLU)

Calculating $z^{(1)} = A^{(1)}x + b^{(1)}$:

$$A^{(1)}x = \begin{pmatrix} 3 \\ -1 \\ 2 \\ 0 \end{pmatrix}, z^{(1)} = \begin{pmatrix} 3 \\ 0 \\ 1 \\ 0 \end{pmatrix}$$

Apply ReLU activation $a^{(1)} = \max(0, z^{(1)})$:

$$a^{(1)} = \begin{pmatrix} 3 \\ 0 \\ 1 \\ 0 \end{pmatrix}$$

Layer 2: Second Hidden Layer (Sigmoid) Calculate $z^{(2)} = A^{(2)}a^{(1)} + b^{(2)}$:

$$A^{(2)}a^{(1)} = \begin{pmatrix} 3 \\ 0 \\ 3 \\ 1 \\ 0 \\ 4 \end{pmatrix}, z^{(2)} = \begin{pmatrix} 3 \\ 0 \\ 2 \\ 1 \\ 0 \\ 4 \end{pmatrix}$$

Apply Sigmoid activation $a^{(2)} = \frac{1}{1+e^{-z^{(2)}}}$:

$$a^{(2)} = \begin{pmatrix} 0.95 \\ 0.50 \\ 0.88 \\ 0.73 \\ 0.50 \\ 0.98 \end{pmatrix}$$

Layer 3: Output Layer (Softmax) Calculate $z^{(3)} = A^{(3)}a^{(2)} + b^{(3)}$:

$$A^{(3)}a^{(2)} = \begin{pmatrix} 0.95 \\ 0.50 \\ 0.88 \\ 0.73 \\ 1.83 \end{pmatrix}, z^{(3)} = \begin{pmatrix} 0.95 \\ 0.50 \\ 0.88 \\ 0.73 \\ 1.83 \end{pmatrix}$$

Apply Softmax activation $a_i^{(3)} = \frac{e^{z_i^{(3)}}}{\sum e^{z_j^{(3)}}}$:

$$a^{(3)} = \begin{pmatrix} 0.17 \\ 0.11 \\ 0.16 \\ 0.14 \\ 0.42 \end{pmatrix}$$

$a^{(3)}$ is the final output vector.

2. Represent NN as composite function

The complete composite function is given by applying the transformations sequentially:

$$f(x) = \text{softmax}(A^{(3)}\sigma(A^{(2)}\text{ReLU}(A^{(1)}x + b^{(1)}) + b^{(2)}) + b^{(3)})$$